

# SF2740 Graph Theory

## Review of the paper

### “Brooks’ theorem on powers of graphs” [BB14]

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## 1 Introduction

The reviewed paper, “Brooks’ theorem on powers of graphs” [BB14], gives a generalisation of Brooks’ theorem on choosability of graphs to powers of graphs. In particular, with the notation introduced below, M. Bonamy and N. Bousquet show that the graph  $G^k$  is  $(D(k, \Delta) - 1)$ -choosable for  $k \geq 3$ . In the following, we consider  $G$  to be a simple, connected graph with maximum degree  $\Delta \geq 3$ .

Together with the notion of colourability studied in the course, the authors consider two generalisations. A graph  $G$  is said to be  **$k$ -choosable** if for every assignment of a list of  $k$  colours to *each* vertex of  $G$ , there is a way to colour the vertices of  $G$  so that adjacent vertices are assigned different colours. This property is also called *list colouring*. Meanwhile, it is said to be  **$k$ -paintable** if, in addition, such colouring can be achieved by means of an algorithm that, no matter the order of the colours, considers at each step  $i$  the vertices that contain the  $i$ -th colour in their lists, and decides which vertices to colour with it. A painted vertex cannot be repainted afterwards. This property is also called *online list colouring*. It is easy to check that a  $k$ -paintable graph is  $k$ -choosable, and a  $k$ -choosable graph is  $k$ -colourable.

Once these concepts are defined, the authors start by reminding the choosability version of Brooks’ theorem.

**Theorem 1.1** (Brooks, Theorem 1 [BB14]). *Except for cliques, every graph is  $\Delta$ -choosable.*

The clique case is always trivial so we assume that  $G$  is not a clique in the following. The next step is to consider powers of graphs. The  $k$ -th power  $G^k$  of  $G = (V, E)$  is defined as the graph with same vertex set  $V$  and such that two vertices are connected if the distance between them in  $G$  is at most  $k$ . Since  $\Delta(G^2) \leq \Delta^2$ , in general  $G^2$  is  $\Delta^2$ -choosable. For subcubic graphs ( $\Delta = 3$ ), the following improvement can be done.

**Theorem 1.2** (Theorem 2 [BB14]). *Except for the Petersen graph, the square of any subcubic graph is 8-choosable.*

This result was proved by Cranston and Kim [CK08], who furthermore conjectured its generalisation for  $\Delta > 3$ , except for Moore graphs. This extends Theorem 1.2, given that the Petersen graph is the only Moore graph with  $\Delta = 3$ . Additionally, some generalisations of this statement for higher powers of graphs were conjectured, as we state next. We note that the greatest possible maximum degree of  $G^k$  can be expressed as

$$D(k, \Delta) := \Delta \frac{(\Delta - 1)^k - 1}{\Delta - 2}.$$

In particular  $D(2, \Delta) = \Delta^2$ , and hence the following conjecture extends the statement of Theorem 1.2.

**Conjecture 1.3** (Conjecture 4 [BB14]). *For any  $k \in \mathbb{N}^*$ , except for Moore graphs when  $k = 2$ ,  $G^k$  is  $(D(k, \Delta) - 1)$ -choosable.*

In the reviewed paper, the authors give a proof for  $k \geq 3$ .

**Theorem 1.4** (Theorem 5 [BB14]). *For  $k \geq 3$ , the  $k$ -th power of any graph is  $(D(k, \Delta) - 1)$ -choosable.*

The case  $k = 2$  was independently proved by Cranston and Rabern [CR16], completing the proof of Conjecture 1.3. The authors leave the following generalisation as future work.

**Conjecture 1.5** (Conjecture 7 [BB14]). *For any  $k \in \mathbb{N}^*$ , except for a finite number of graphs, the  $k$ -th power of any graph is  $(D(k, \Delta) + 1 - k)$ -choosable.*

## 2 Results

We state the central results of the article and sketch the most relevant proofs. Lemma 2.1 gives a set of conditions for which the main theorem can be proven more easily.

**Lemma 2.1** (Lemma 8 [BB14]). *Given a graph  $G$ , the power graph  $G^k$  is  $(D(k, \Delta) - 1)$ -paintable if  $G$  satisfies any of the following:*

- (i)  $G$  contains a vertex of degree smaller than  $\Delta$ .
- (ii)  $G$  contains a cycle shorter than  $2k$ .
- (iii)  $G$  contains two intersecting cycles of length  $2k$ .
- (iv)  $\text{diam}(G) \leq k$ .

So now we may assume that  $G$  is  $\Delta$ -regular, contains only cycles of length at least  $2k$ , does not contain intersecting cycles of length  $2k$  and has two vertices at distance at least  $k + 1$  from each other. Lemma 2.2 proves the existence of four specific vertices which are crucial for the proof of the main theorem. We briefly describe the proof.

**Lemma 2.2** (Lemma 9 [BB14]). *With the assumptions above, the graph  $G$  contains two vertices  $x_1, y_1$  at distance  $k + 1$  from each other, with two respective neighbours  $x_2, y_2$  at distance at least  $k + 1$  from each other.*

*Proof (sketch).* The existence of  $x_1, y_1$  follows from Lemma 2.1 (iv). Assuming that every two neighbours of  $x_1, y_1$  are at distance at most  $k$  from each other, the pigeonhole principle and again Lemma 2.1 allows us to find a cycle of length 4, which is a contradiction to the assumptions.  $\square$

The main result in the paper is actually a stronger version of Theorem 1.4. The assumptions on  $G$  are the same as in Lemma 2.2.

**Theorem 2.3** (Theorem 6 [BB14]). *For  $k \geq 3$ , the  $k$ -th power of any graph is  $(D(k, \Delta) - 1)$ -paintable.*

*Proof (sketch).* Let  $M := D(k, \Delta)$ . An algorithm is described to online list colour the graph  $G^k$  given a list  $L(u)$  of  $M - 1$  colours for each vertex  $u \in G$ .

Let  $x_1, y_1, x_2, y_2$  be the vertices obtained in Lemma 2.2. Then we can find two vertices  $v, w$  in a path  $xPy$  such that  $v$  is a neighbour of all  $x_1, y_1, x_2, y_2$  in  $G^k$ , while  $w$  is a neighbour of at least  $x_1, y_1$ .

We order the set  $V(G)$  as follows: first  $x_1, y_1, x_2, y_2$ , then all other vertices in decreasing distance to  $\{v, w\}$ , breaking ties arbitrarily and finishing with  $w, v$ . Given an ordering of all the colours in the lists, the colouring algorithm consists in, for each color  $c_i$ , treating the vertices  $x_1, y_1, x_2, y_2$  first, and then considering the remaining vertices in the stated order. For each of these vertices  $u$ , we colour it if  $c_i \in L(u)$  and none of its neighbours in  $G^k$  is coloured with  $c_i$ .

The number of *constraints* of a vertex  $u$  at any step of the algorithm is the number of colours of  $L(u)$  that have already been used to colour neighbours of  $u$  in  $G^k$ . The goal of the algorithm is that every vertex has at most  $M - 2$  constraints at every step, meaning that it can be coloured as  $|L(u)| = M - 1$ .

The authors first show that given an uncoloured vertex  $u \notin \{x_1, y_1, x_2, y_2\}$ , the number of constraints for  $u$  is at most  $M - r(u)$ , where  $r(u)$  is the number of neighbours of  $u$  in  $G^k$  that come after  $u$  in the order. Note that neighbours of  $u$  that come after  $u$  in the order may be coloured before  $u$ , but in any case they do not add a constraint. From this claim it follows that at any step of the algorithm, any vertex  $u \notin \{v, w\}$  has at most  $M - 2$  constraints, and hence it is eventually coloured with a certain  $c_i$ .

It remains to show that  $x_1, y_1, x_2, y_2$  can be coloured in such a way that allows to colour  $v, w$ . We wish that the set  $\{x_1, y_1\}$  imposes at most one constraint on  $v$  and  $w$ , and that the set  $\{x_2, y_2\}$  imposes at most one constraint on  $v$ . This allows  $v$  to have  $M - 2$  constraints, while it can be checked that  $w$  has  $M - 1 - r(w) = M - 2$ , since  $v$  does not add a constraint to  $w$ .

The authors give a list of possible choices for colouring  $x_1, y_1$  and  $x_2, y_2$ . Whenever it is possible to colour each pair in the same colour it is done.

Otherwise, a vertex in the set  $\{x_i, y_i\}$  is coloured if by doing it, the number of constraints that  $\{x_i, y_i\}$  implies on  $v$  or  $w$  is no longer maximum. If this is never the case, the corresponding vertex is coloured in the last possible step. Finally, it is shown that these instructions ensure that all four vertices are coloured and  $v, w$  have at most  $M - 2$  constraints at any step.  $\square$

### 3 Personal assessment

The article provides a generalisation of Brooks' theorem for list colouring powers of graphs. In fact, the authors prove a stronger result (Theorem 2.3), which holds for the property of online list colouring. In turn, this implies Conjecture 1.3 for all integers  $k \geq 3$ . Together with [CR16], this completes the proof of Conjecture 1.3. Moreover, the authors construct an explicit algorithm to list colour  $G^k$  given any assignment of lists of  $D(k, \Delta) - 1$  colours at each vertex.

The interest of the article is twofold. First, it provides an upper bound of the chromatic number of the power of any graph. Powers of graphs are relevant, for instance, in order to study properties that hold in a neighbourhood of a vertex. Secondly, the algorithm constructed in the proof of Theorem 2.3 allows to online list colour  $G^k$  in polynomial time. This can be achieved since once a vertex is coloured, its colour is fixed and the vertex is not revisited. Thus, the amount of steps of the algorithm is bounded by the total number of colours. In particular, the algorithm is also valid for standard colouring  $G^k$ . Nonetheless, its application might not be too interesting in some cases, especially when  $\chi(G)$  is considerably smaller than  $D(k, \Delta)$ . In any case, the authors should probably have included this analysis in the publication.

On a negative side, we found the article somewhat difficult to read. The authors use different terms referring to the same concepts, especially when stating the basic definitions, and do not properly clarify others (e.g. does  $\mathbb{N}^*$  contain 1?). This made it confusing to understand at first sight, and we needed to double-check with other sources. Together with this, some parts of the text feel heavy to read, mainly because of the massive use of parentheses. The structure is also a bit unclear, with the proof of Theorem 2.3 going on for a couple of pages while some partial results are inserted in between, leaving the reader at times guessing what is being proven. Furthermore, basic figures might have helped to illustrate the proof.

In addition, we found a couple of typos in the text. One of them is just a notation issue, since the authors write  $|M|$  as the cardinal of a list, while  $M$  itself already represented this number. In contrast, in the last part of the proof, the inequality  $|L(x_2) \cap L(x_1)| \leq 5$  holds actually also for 4. While this difference is not crucial in their argument, they actually use the bound 4 in the next step, which can be really misleading. Overall, we believe that these details should have been checked.

## References

- [BB14] Marthe Bonamy and Nicolas Bousquet. “Brooks’ theorem on powers of graphs”. In: *Discrete Mathematics* 325 (2014), pp. 12–16.
- [CK08] Daniel W. Cranston and Seog-Jin Kim. “List-coloring the square of a subcubic graph”. In: *Journal of Graph Theory* 57.1 (2008), pp. 65–87.
- [CR16] Daniel W. Cranston and Landon Rabern. “Painting Squares in  $\Delta^2 - 1$  Shades”. In: *The Electronic Journal of Combinatorics* 23 (2016).